

SOP for Impulse Traveling Head Study

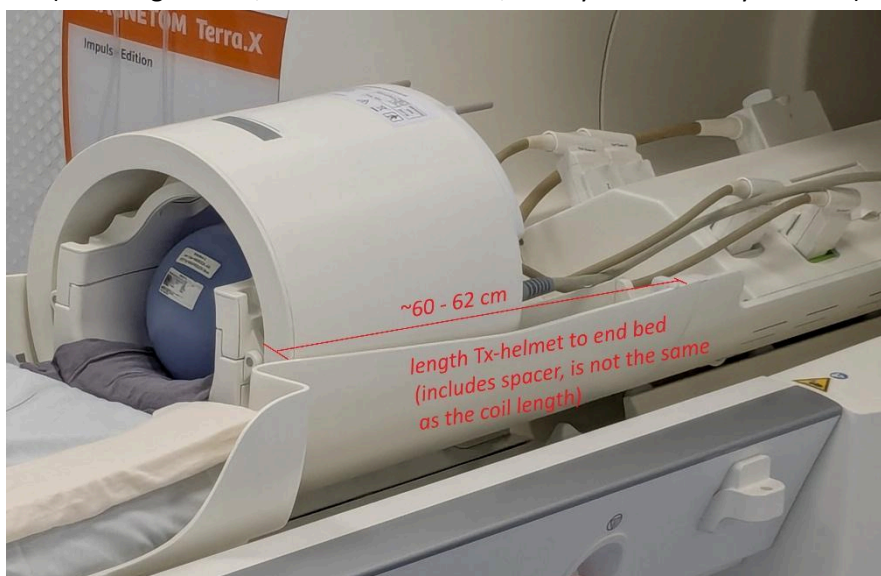
General comments:

- Feedback please at: hendrik.mattern@ovgu.de & saskia.bollmann@ug.edu.au
- includes **fMRI with task** (psychopy)
- acquired **without physio data**
- use the following pillows for the head (see image below):
 - Star-shaped pillow to lie on
 - Triangle-shaped pillows over the ears



Document setting:

- Volunteer name: _____
- Volunteer weight: ____ kg
- Volunteer height: ____ cm
- Operator name: _____
- Scan date: _____
- Distance end patient bed to end transmit coil: _____ cm
(see image below, 61.5 cm in Stanford, ideally measured by Hendrik)



Notes on PsychoPy:

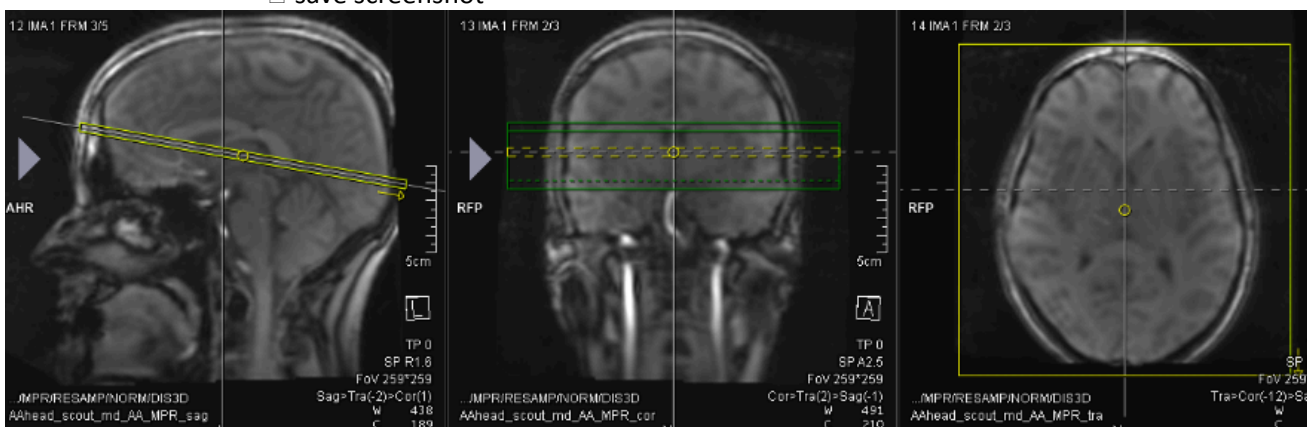
- There is a dedicated docx explaining how to open and run the tasks

Notes on FOV planning:

- Please obey the following rules and adhere to them at all times when planning the slice/FOV positioning (shifts & angulation):
 - Not allowed when labeled as **NO PLANNING**
 - Only allowed when the AutoAlign obviously failed; labeled as **DOUBLECHECKING**
 - Has to be done manually when labeled as **MANUAL PLANNING**
 - Done via **COPYREFERENCE**

Scanning:

1. **Localizer** [strict SAG & in ISO] **NO PLANNING**
2. **AAhead_scout_md** [strict SAG & in ISO] **NO PLANNING**
 - ☐ FYI: used for autoalign, back of the brain might be missing, use localizer if in doubt when planning manually
3. **Manual adjustments for B0 map etc.**
 - FWHM before shimming: _____ Hz
 - Run shim (2x GRE Brain with measure, calculate, apply)
 - FWHM after shimming: _____ Hz
4. **gre_b0map_4iso_sag** [strict SAG & in ISO] **NO PLANNING**
5. **dzne_b1map_5iso_sag** [strict SAG & in ISO] **NO PLANNING**
 - ☐ copy adjustment volume from gre_b0map_4iso_sag
6. **gre_FA10_acpc** [ACPC via AutoAlign; screenshot only for **DOUBLECHECKING**]
 - Reference voltage: _____ V
 - ☐ save screenshot



7. **gre_FA0_acpc** [ACPC via **COPYREFERENCE**]
 - ☐ doublecheck RefVoltage = 0V
 - ☐ confirm RefVoltage when asked



gre_FA10_up [corpus callosum, **MANUAL PLANNING**]

- ☐ use ACPC angulation by copying slice position from gre_FA10_acpc
- ☐ click “...” behind position and add 20 mm to the “H” field (FYI: if “F” subtract 20 mm)
- ☐ save screenshot of FOV planning

8. **gre_FA0_up** [corpus callosum, via **COPYREFERENCE**]

- ☐ copy slice position from previous sequence
- ☐ doublecheck RefVoltage = 0V
- ☐ confirm RefVoltage when asked

9. **mp2rage_0p7iso_patientSpecific** [strict SAG & in ISO, **NO PLANNING**]

- ☐ if SAR, switch to first level

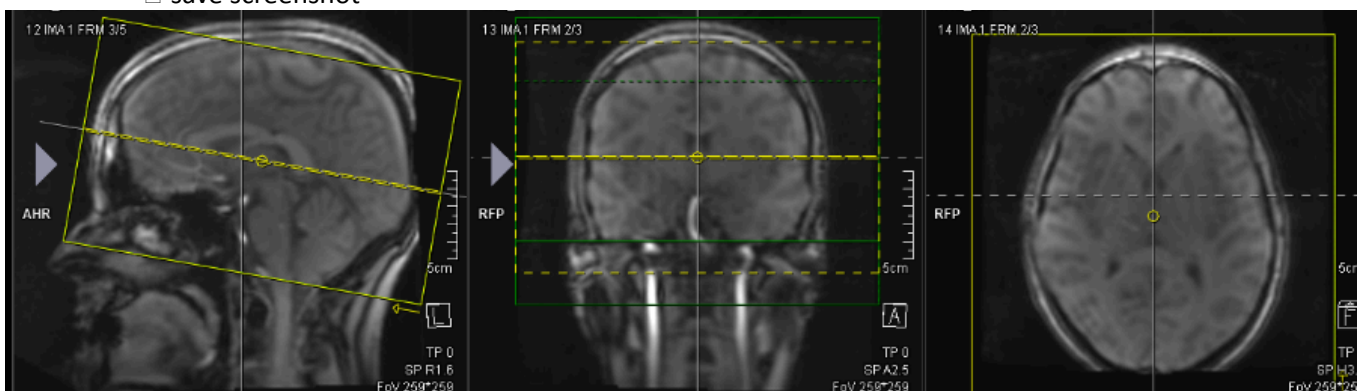
10. **mp2rage_0p7iso_TrueForm** [strict SAG & in ISO, **NO PLANNING**]

11. **Manual adjustments for diffusion**

- FWHM before shimming: _____ Hz
- Run shim (2x GRE Brain with measure, calculate, apply)
- FWHM after shimming: _____ Hz

12. **ep2d_diff_1p3iso_b2k5_PA** [ACPC via AutoAlign; screenshot only for **DOUBLECHECKING**]

- ☐ save screenshot

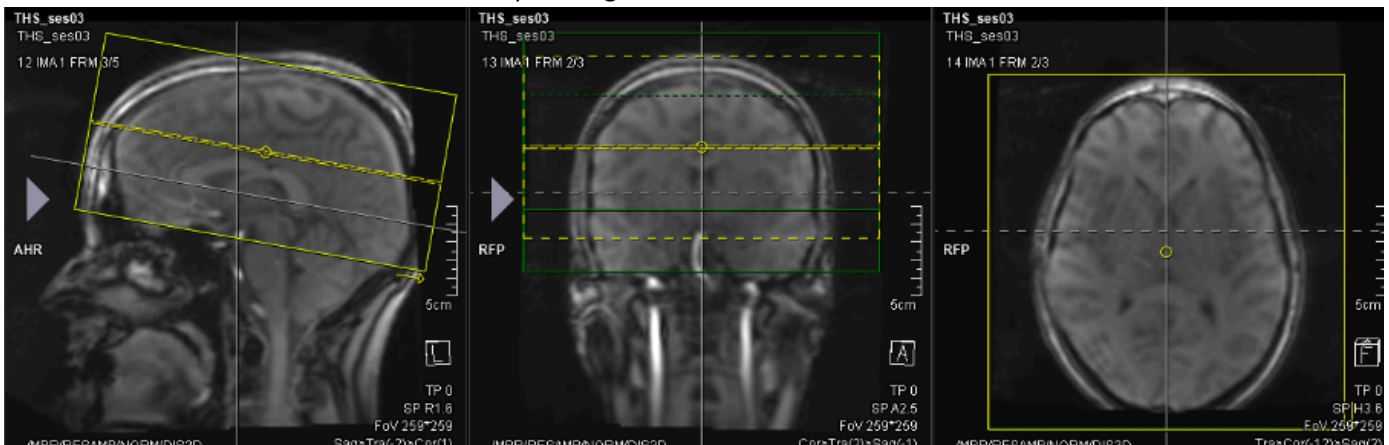


13. **ep2d_diff_1p3iso_b1k0_PA** [ACPC via COPYREFERENCE]
☐ copy adjustment volume from ep2d_diff_1p3iso_b2k5_PA

14. **ep2d_diff_1p3iso_AP** [ACPC via COPYREFERENCE]
☐ copy slice position from previous sequence
→ DOUBLE CHECK phase encoding is "AP"
☐ copy adjustment volume from ep2d_diff_1p3iso_b2k5_PA

15. **Manual adjustments for ep2d**
- FWHM before shimming: _____ Hz
 - Run shim (2x GRE Brain with measure, calculate, apply)
 - FWHM after shimming: _____ Hz

16. **ep2d_bold_1p4mm with TASK** [ACPC via AutoAlign; might need shift up; screenshot only for DOUBLECHECKING]
☐ save screenshot of FOV planning



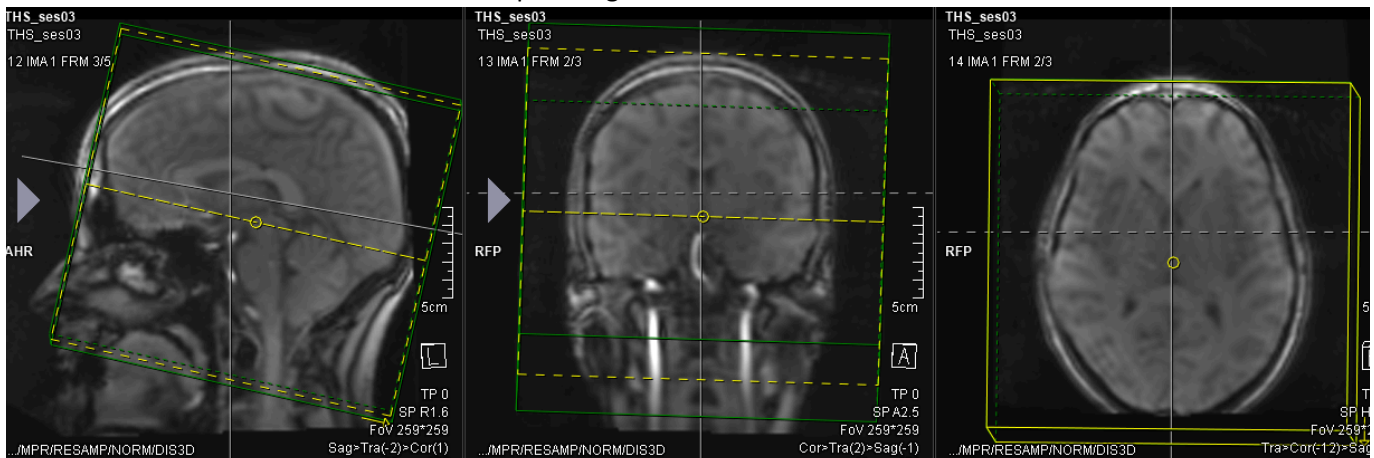
- ☐ FYI: in Stanford the FOV had to be shifted headwards to cover the entire cerebrum
☐ start THS_ep2d_TappingWithTrTiming_Movie_d3.py (see appendix for info)

17. **Manual adjustments for ep3d_vasoNIH**
- FWHM before shimming: _____ Hz
 - Run shim (2x GRE Brain with measure, calculate, apply)
 - FWHM after shimming: _____ Hz

→ see next page

18. **dzne_ep3d_vasoNIH with Task, run 1** [ACPC whole brain, A>>P, might need shift down; screenshot only for **DOUBLECHECKING**]

☐ save screenshot of FOV planning



☐ start THS_ep3d_TappingWithTrTiming_Movie_d3.py

19. **dzne_ep3d_vasoNIH with Task, run 2** [ACPC, whole brain via **COPYREFERENCE**]

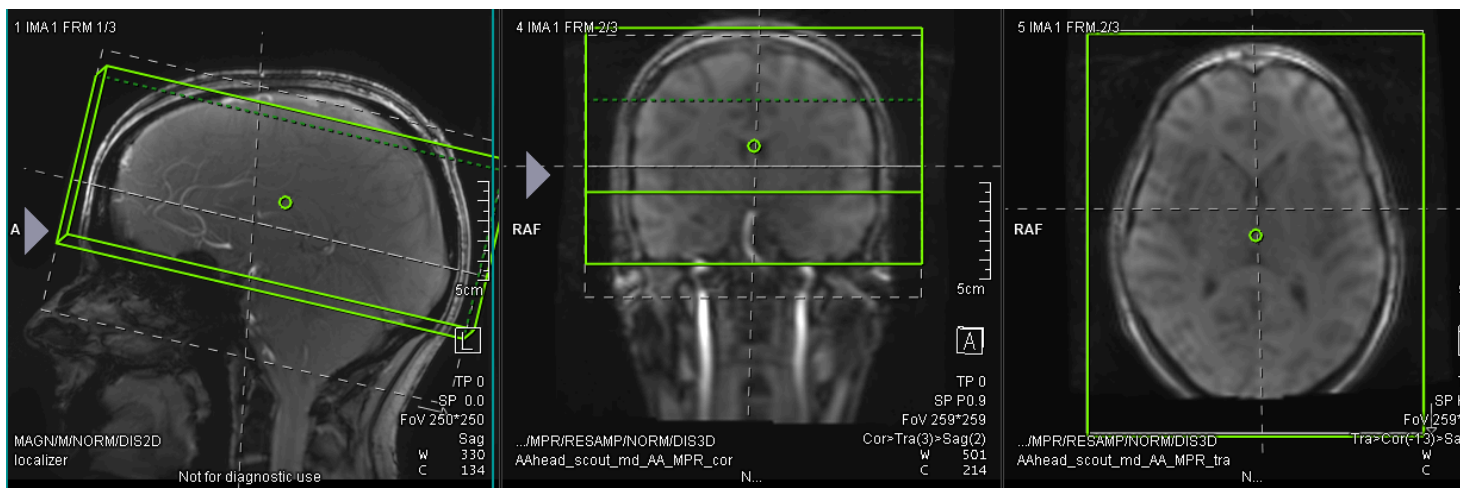
☐ start THS_ep3d_TappingWithTrTiming_Movie_d3.py

20. Manual adjustments for meso-vein Frankfurt

- FIRST PLAN FOV & ADJUSTMENT VOLUME (see notes below)
- FWHM before shimming: _____ Hz
- Run shim (2x GRE Brain with measure, calculate, apply)
- FWHM after shimming: _____ Hz

21. **dzne_ep3d_Frankfurt_6echoes** [ACPC, A>>P, smaller Adj.Vol, **MANUAL PLANNING**]

☐ save screenshot of FOV planning

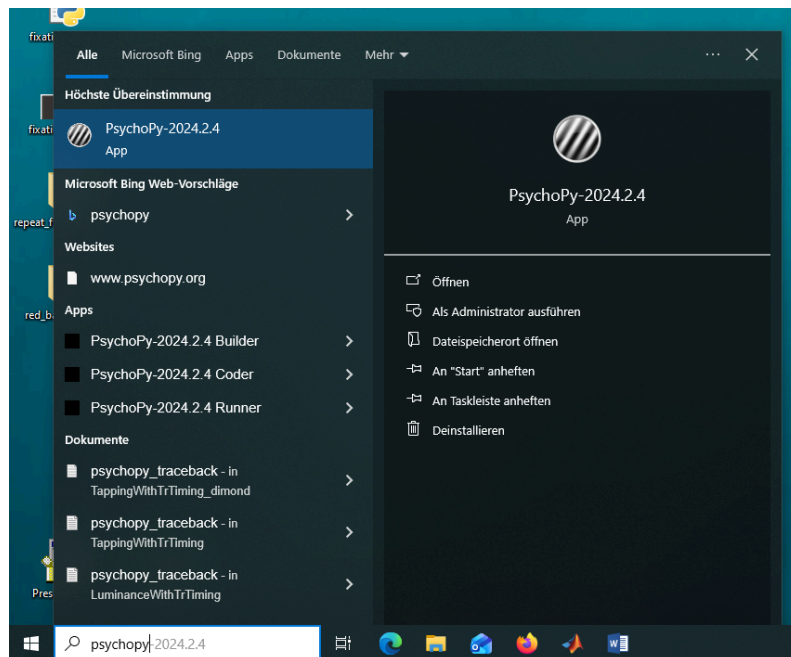


- ☐ ACPC auto align, typically no adjustment of FOV required
- ☐ IMPORTANT: Adjustment volume is smaller than the FOV but has the same angulation.
- ☐ Adjustment volume WITHOUT EYES and covers most of the cerebrum
- ☐ FOV covers entire cerebrum and parts of cerebellum
- ☐ FYI: recon tasks ~15 min

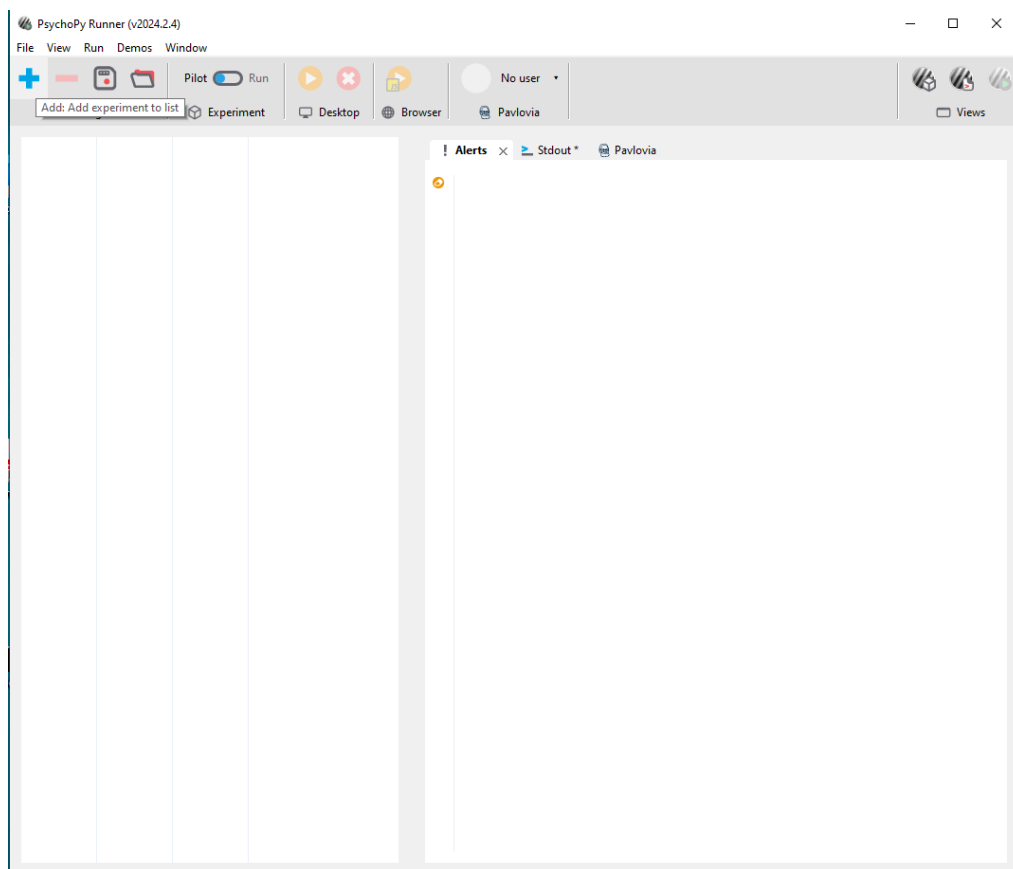
Appendix: How to start the paradigm in PsychoPy?

FYI: original code for THS here https://github.com/layerfMRI/Phychopy_git

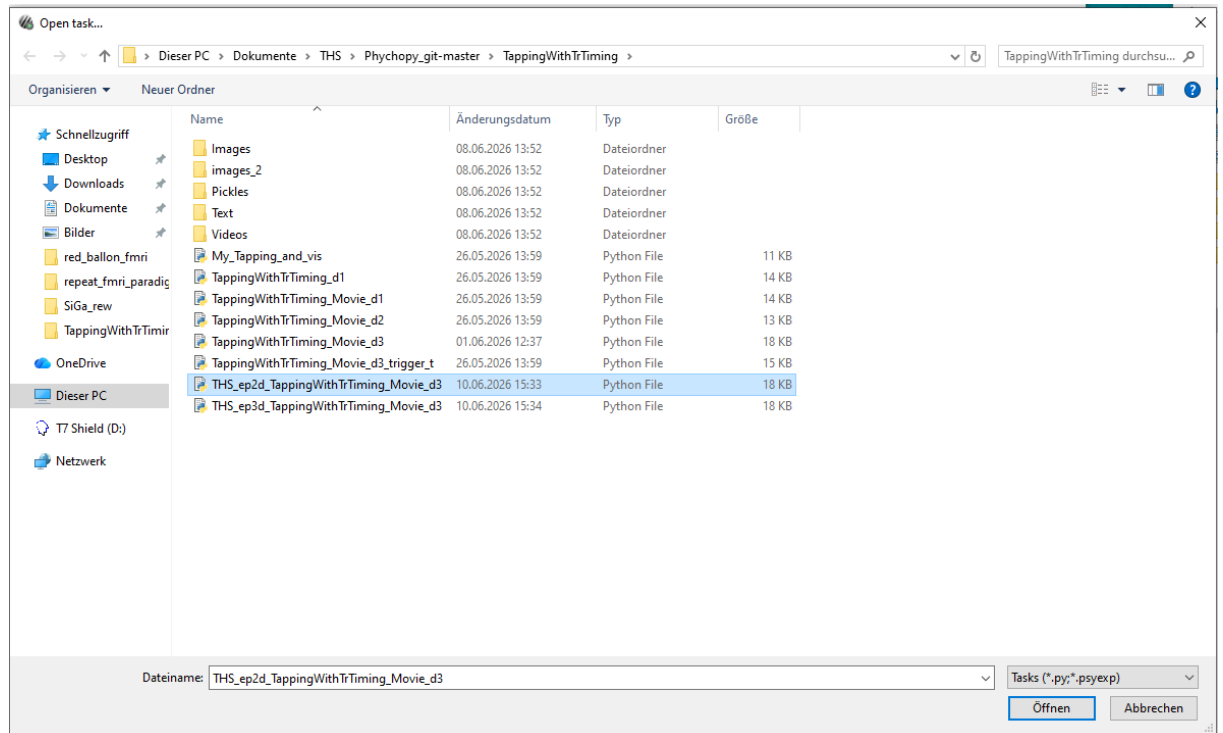
1.) Open PsychoPy



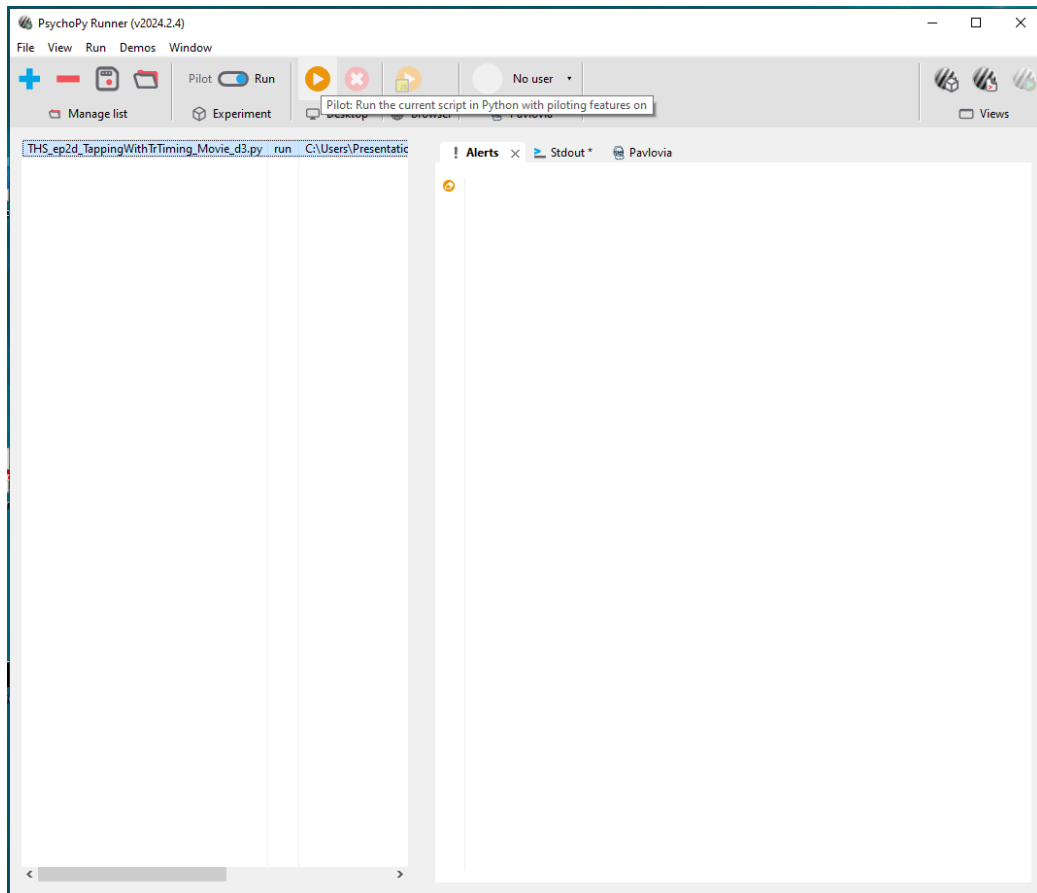
2.) Click "+" to add an experiment



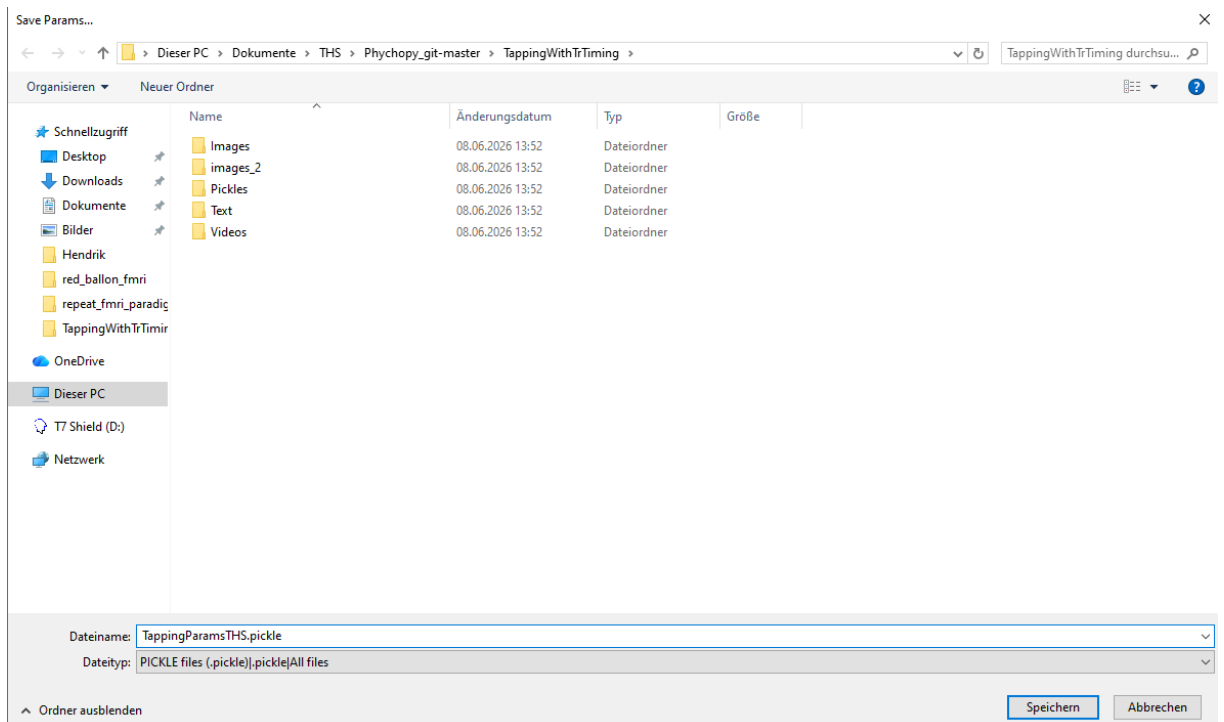
3.) Select file for task



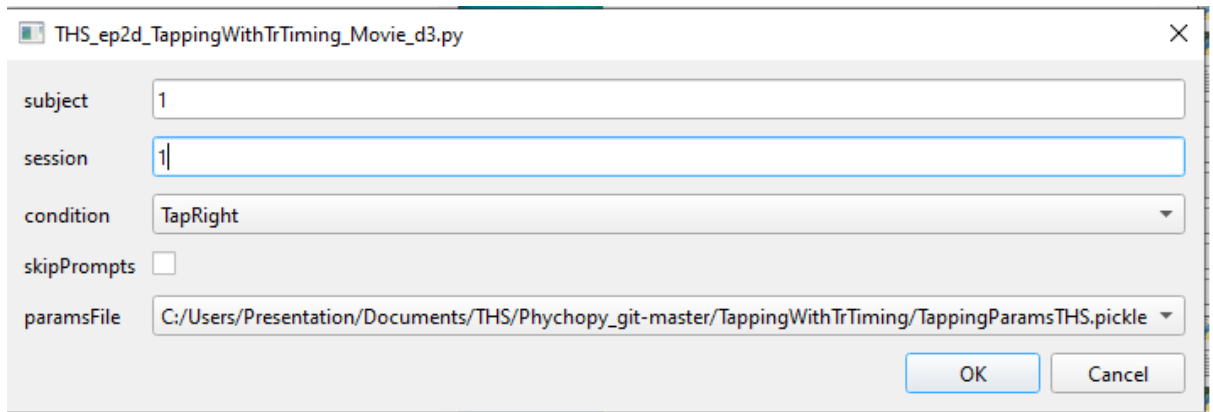
- 4.) Run the task by clicking the orange “play” button in the main PsychoPy window
IMPORTANT: the file you want to run needs to be selected/highlighted in blue



- 5.) Name the Log-File (PICKLE format) in the format “THS_<scan_date>_ep2d_run<Nr.>”, or “THS_<scan_date>_ep3d_run<Nr.>”, e.g. THS_2026_06_10_ep3d_run2 for a 3D EPI on the 10.06.2026, second run



- 6.) Enter subject ID=1, and session number=1 as each EPI will have an independent Log-File/PICKLE



The screenshot shows a Python script window titled "THS_ep2d_TappingWithTrTiming_Movie_d3.py". It contains several input fields and a checkbox:

- subject**: A text box containing the value "1".
- session**: A text box containing the value "1".
- condition**: A dropdown menu with "TapRight" selected.
- skipPrompts**: A checkbox that is currently unchecked.
- paramsFile**: A text box containing the path "C:/Users/Presentation/Documents/THS/Phychopy_git-master/TappingWithTrTiming/TappingParamsTHS.pickle".

At the bottom right of the window are two buttons: "OK" and "Cancel".

- 7.) Press "Esc" or "q" twice to end the measurement if required